

# Circular Economy Ontology Network (CEON) - Process Module

## Metadata

### IRI

<http://w3id.org/CEON/ontology/process/>

### Title

Circular Economy Ontology Network (CEON) - Process Module

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### License

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### Version Iri

<http://w3id.org/CEON/ontology/process/0.1/>

### Version Info

0.1

### Preferred Namespace Prefix

process

### Preferred Namespace Uri

<https://w3id.org/CEON/ontology/process/>

### Description

The Process module of CEON (Circular Economy Ontology Network).

### Covers Requirements

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESIDE D3.1: T8-2

## Classes

### Assembling Process<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/process/AssemblingProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

### Buy Resource Process<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/process/BuyResourceProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

## Co2Emission<sup>C</sup>

|                            |                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/CO2Emission">http://w3id.org/CE0N/ontology/process/CO2Emission</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">resourceODP:Resource<sup>C</sup></a>                                                                  |
| <b><u>In Range Of</u></b>  | <a href="#">producesCO2<sup>Op</sup></a>                                                                          |

## Cathalyst<sup>C</sup>

|                            |                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/Cathalyst">http://w3id.org/CE0N/ontology/process/Cathalyst</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">resourceODP:Resource<sup>C</sup></a>                                                              |
| <b><u>In Range Of</u></b>  | <a href="#">usesCathalyst<sup>Op</sup></a>                                                                    |

## Change Resource Process<sup>C</sup>

|                            |                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/ChangeResourceProcesses">http://w3id.org/CE0N/ontology/process/ChangeResourceProcesses</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">processODP:Transformation<sup>C</sup></a>                                                                                     |

## Contact Process<sup>C</sup>

|                            |                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/ContactProcess">http://w3id.org/CE0N/ontology/process/ContactProcess</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">processODP:Transformation<sup>C</sup></a>                                                                   |

## Deconstruction Process<sup>C</sup>

|                            |                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/DeconstructionProcesses">http://w3id.org/CE0N/ontology/process/DeconstructionProcesses</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">processODP:Transformation<sup>C</sup></a>                                                                                     |

## Dismantle Process<sup>C</sup>

|                            |                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/DismantleProcess">http://w3id.org/CE0N/ontology/process/DismantleProcess</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">processODP:Transformation<sup>C</sup></a>                                                                       |

## Energy<sup>C</sup>

|                            |                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/Energy">http://w3id.org/CE0N/ontology/process/Energy</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">resourceODP:Resource<sup>C</sup></a>                                                        |
| <b><u>In Range Of</u></b>  | <a href="#">needsEnergy<sup>Op</sup></a>                                                                |

## Ensure Claim Process<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/EnsureClaimProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

## Issuing Certificate Process<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/IssuingCertificateProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

## Manufacturing Process<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/ManufacturingProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

## Offset Process<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/OffsetProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

## Production Process<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/ProductionProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

## Recycle Process<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/RecycleProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

## Refurbishment Process<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/RefurbishmentProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

## Remove Process<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/RemoveProcess>

**Sub Class Of** [processODP:Transformation<sup>C</sup>](#)

### Remove Resource Process<sup>C</sup>

|                            |                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/RemoveResourceProcesses">http://w3id.org/CE0N/ontology/process/RemoveResourceProcesses</a> |
| <b><u>Sub Class Of</u></b> | <a href="#"><u>processODP:Transformation</u></a> <sup>C</sup>                                                                             |

### Repair Process<sup>C</sup>

|                            |                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/RepairProcess">http://w3id.org/CE0N/ontology/process/RepairProcess</a> |
| <b><u>Sub Class Of</u></b> | <a href="#"><u>processODP:Transformation</u></a> <sup>C</sup>                                                         |

### Resell Process<sup>C</sup>

|                            |                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/ResellProcess">http://w3id.org/CE0N/ontology/process/ResellProcess</a> |
| <b><u>Sub Class Of</u></b> | <a href="#"><u>processODP:Transformation</u></a> <sup>C</sup>                                                         |

### Reuse Process<sup>C</sup>

|                            |                                                                                                                     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/ReuseProcess">http://w3id.org/CE0N/ontology/process/ReuseProcess</a> |
| <b><u>Sub Class Of</u></b> | <a href="#"><u>processODP:Transformation</u></a> <sup>C</sup>                                                       |

### Sell Resource Process<sup>C</sup>

|                            |                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/SellResourceProcess">http://w3id.org/CE0N/ontology/process/SellResourceProcess</a> |
| <b><u>Sub Class Of</u></b> | <a href="#"><u>processODP:Transformation</u></a> <sup>C</sup>                                                                     |

### Service Process<sup>C</sup>

|                            |                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/ServiceProcess">http://w3id.org/CE0N/ontology/process/ServiceProcess</a> |
| <b><u>Sub Class Of</u></b> | <a href="#"><u>processODP:Transformation</u></a> <sup>C</sup>                                                           |

### Share Resource Process<sup>C</sup>

|                            |                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/ShareResourceProcess">http://w3id.org/CE0N/ontology/process/ShareResourceProcess</a> |
| <b><u>Sub Class Of</u></b> | <a href="#"><u>processODP:Transformation</u></a> <sup>C</sup>                                                                       |

### Take Back Process<sup>C</sup>

|                            |                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/process/TakeBackProcess">http://w3id.org/CE0N/ontology/process/TakeBackProcess</a> |
| <b><u>Sub Class Of</u></b> | <a href="#"><u>processODP:Transformation</u></a> <sup>C</sup>                                                             |

## Transformation<sup>c</sup>

**IRI** <http://w3id.org/CE0N/ontology/processODP/Transformation>

### Super Class Of

[AssemblingProcess<sup>c</sup>](#)  
[BuyResourceProcess<sup>c</sup>](#)  
[ChangeResourceProcess<sup>c</sup>](#)  
[ContactProcess<sup>c</sup>](#)  
[DeconstructionProcess<sup>c</sup>](#)  
[DismantleProcess<sup>c</sup>](#)  
[EnsureClaimProcess<sup>c</sup>](#)  
[IssuingCertificateProcess<sup>c</sup>](#)  
[ManufacturingProcess<sup>c</sup>](#)  
[OffsetProcess<sup>c</sup>](#)  
[ProductionProcess<sup>c</sup>](#)  
[RecycleProcess<sup>c</sup>](#)  
[RefurbishmentProcess<sup>c</sup>](#)  
[RemoveProcess<sup>c</sup>](#)  
[RemoveResourceProcess<sup>c</sup>](#)  
[RepairProcess<sup>c</sup>](#)  
[ResellProcess<sup>c</sup>](#)  
[ReuseProcess<sup>c</sup>](#)  
[SellResourceProcess<sup>c</sup>](#)  
[ServiceProcess<sup>c</sup>](#)  
[ShareResourceProcess<sup>c</sup>](#)  
[TakeBackProcess<sup>c</sup>](#)

## Resource<sup>c</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/Resource>

**In Range Of** [resultingProduct<sup>op</sup>](#)

### Super Class Of

[CO2Emission<sup>c</sup>](#)  
[Cathalyst<sup>c</sup>](#)  
[Energy<sup>c</sup>](#)

## Object Properties

### needs energy<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/needsEnergy>

**Sub Property Of** [processODP:hasInput<sup>op</sup>](#)

**Range** [Energy<sup>c</sup>](#)

### produces co2<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/producesCO2>

**Sub Property Of** [processODP:hasOutput<sup>op</sup>](#)

**Range** [CO2Emission<sup>c</sup>](#)

resulting product<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/resultingProduct>

**Sub Property Of** [processODP:hasOutput<sup>op</sup>](#)

**Range** [resourceODP:Resource<sup>c</sup>](#)

uses cathalyst<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/process/usesCathalyst>

**Sub Property Of** [processODP:hasInput<sup>op</sup>](#)

**Range** [Cathalyst<sup>c</sup>](#)

has input<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/processODP/hasInput>

**Super Property Of**

- [needsEnergy<sup>op</sup>](#)
- [usesCathalyst<sup>op</sup>](#)

has output<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/processODP/hasOutput>

**Super Property Of**

- [producesCO2<sup>op</sup>](#)
- [resultingProduct<sup>op</sup>](#)

## Annotation Properties

contributor<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/contributor>

created<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/created>

creator<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/creator>

description<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/description>

|                                          |                                                                                                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| license <sup>ap</sup>                    |                                                                                                                                                                                               |
| IRI                                      | <a href="http://purl.org/dc/terms/license">http://purl.org/dc/terms/license</a>                                                                                                               |
| title <sup>ap</sup>                      |                                                                                                                                                                                               |
| IRI                                      | <a href="http://purl.org/dc/terms/title">http://purl.org/dc/terms/title</a>                                                                                                                   |
| preferred namespace prefix <sup>ap</sup> |                                                                                                                                                                                               |
| IRI                                      | <a href="http://purl.org/vocab/vann/preferredNamespacePrefix">http://purl.org/vocab/vann/preferredNamespacePrefix</a>                                                                         |
| preferred namespace uri <sup>ap</sup>    |                                                                                                                                                                                               |
| IRI                                      | <a href="http://purl.org/vocab/vann/preferredNamespaceUri">http://purl.org/vocab/vann/preferredNamespaceUri</a>                                                                               |
| covers requirements <sup>ap</sup>        |                                                                                                                                                                                               |
| IRI                                      | <a href="http://www.ontologydesignpatterns.org/schemas/cannotation/schema.owl#coversRequirements">http://www.ontologydesignpatterns.org/schemas/cannotation/schema.owl#coversRequirements</a> |

## Namespaces

```

:
    http://w3id.org/CE0N/ontology/process/
dcterms
    http://purl.org/dc/terms/
odp
    http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#
owl
    http://www.w3.org/2002/07/owl#
processODP
    http://w3id.org/CE0N/ontology/processODP/
prov
    http://www.w3.org/ns/prov#
rdf
    http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs
    http://www.w3.org/2000/01/rdf-schema#
resourceODP
    http://w3id.org/CE0N/ontology/resourceODP/
vann
    http://purl.org/vocab/vann/

```

## Legend

|    |                          |
|----|--------------------------|
| c  | Classes                  |
| op | Object<br>Properties     |
| ap | Annotation<br>Properties |